Condensed Novelty Check: Delay–Layout Coupling in UWB Self-Calibration

Executive summary for paper planning — grounded in the Vicon validation report (`main_EN`)

Note for the reader: this one-page summary only positions the novelty of the delay–layout coupling result against the literature; every number is taken from the `main_EN` report you already have. "`v4-io`" is the production self-calibrated anchor layout; the "common-mode" layout is the metric-corrected diagnostic re-parameterization of the same anchors (`main_EN`, § Delay–Layout Coupling).

1 Thesis

UWB anchor self-calibration couples antenna-delay/range-bias parameters with layout scale, so same-environment tag accuracy is not a reliable proxy for physical calibration quality. In the Erlangen campaign the production `v4-io` self-calibrated layout is scale-biased (metric-incorrect): an under-absorbed common-mode range bias expands it to an AutoPos-to-Vicon Sim(3) scale of 0.958 (a range-bias signature, not a true geometric distortion), while a common-mode delay re-parameterization recovers metric scale (1.010) and nearly halves the rigid anchor RMSE (105.4 to 63.0 mm). Yet the metric-correct common-mode layout gives a *worse* same-environment static median (109.5 mm) than the scale-biased `v4-io` layout (72.7 mm) when the tag delay is left uncorrected, because the layout's scale bias compensates an uncorrected positive tag-device delay. Per-frame joint tag-delay estimation restores the common-mode layout to 72.9 mm median but destabilizes the tail (P95 317.5 mm). A scale-biased, metric-incorrect layout can therefore win in-environment *when the tag delay is left uncorrected*—once it is corrected the metric-correct layout matches it (72.9 mm median), so this is a conditional failure mode, not a claim that a metric-incorrect layout is inherently a better positioner. It is direct evidence that metric correctness and positioning accuracy must be validated separately.

2 Literature Gap

What is known	What is NOT known (our gap)
UWB self-calibration estimates anchors from radio ranges, robot motion, multihop links, or auxiliary sensing (Hamer & D'Andrea, 2018; Corbalan et al., 2023; Van Herbruggen et al., 2023; Almansa et al., 2020; Ledergerber et al., 2015; Ridolfi et al., 2021; Shi et al., 2019; Piavanini et al., 2022; Queralta et al., 2020).	These papers do not isolate whether the layout with the best tag accuracy is metrically correct, nor whether tag accuracy can rank a scale-biased layout above a validated metric one.
Antenna-delay and range-bias calibration are established UWB problems (Ledergerber & D'Andrea, 2018; De Preter et al., 2019; Shalaby et al., 2023; Liu et al., 2024; Shah et al., 2021, 2022).	In UWB ranging, delay/range-bias is usually a nuisance correction (in acoustic/TOA self-calibration it is a first-class jointly-solved unknown); its ability to absorb global layout-scale error in-environment is not characterized.
NLOS mitigation uses channel statistics, learned models, robust weighting, or biased-range models (Wymeersch et al., 2012; Di Franco et al., 2017; Marano et al., 2010; Guvenc et al., 2007; Venkatesh & Buehrer, 2007; Meghani et al., 2019; Jeong et al., 2023; Kong, 2023; Sung et al., 2023; Yang et al., 2024).	Structured positive range bias is not connected to which self-calibration geometry is physically valid.
Fisher/CRLB, rigidity, and observability analyses describe weak directions and rank deficiencies in localization and calibration networks (Shen et al., 2010; Aspnes et al., 2006; Patwari et al., 2005; Wymeersch et al., 2009; Prorok, 2013; Jourdan et al., 2008; Hesch et al., 2014; Goudar et al., 2021).	Prior work does not connect a coupled delay–scale direction to a measured scale-bias cancellation result (a scale-biased layout winning in-environment) in UWB self-calibration.
Scale–propagation-parameter ambiguity is known in acoustic self-calibration and related ranging systems (Burgess et al., 2015). AniTrack reported self-localized anchors outperforming surveyed anchors in tag accuracy (13.96 vs 16.57 cm, $n = 7$) (Luder et al., 2025).	The interaction with structured UWB range bias, and the resulting in-environment advantage of the scale-biased layout, is not reported. A consistent external observation (within one SD, competing explanations not excluded), not a diagnosis: no Sim(3) scale, delay–layout coupling, or causal intervention.

3 Our Contribution

The first contribution is scientific: we quantitatively characterize delay–layout coupling in UWB anchor self-calibration. A four-way ablation (self-calibrated vs Vicon-measured anchor coordinates $\times$ residual-correction treatments) shows the fitted residual corrections are frame-conditioned: Vicon-measured anchor coordinates degrade tag localization (311 mm RMSE with no correction, 252 mm with transplanted corrections) unless the corrections are re-estimated in the Vicon frame (78 mm). A common-mode delay re-parameterization ($d_i = c + e_i$, $c = 112$ mm) recovers metric scale from inter-anchor ranges alone (Sim(3) 0.958 $\rightarrow$ 1.010; rigid RMSE 105.4 $\rightarrow$ 63.0 mm); clamping the common mode to zero raises the solver cost by about 70%, and five perturbed initializations converge to the same scale, indicating a weakly observable (not strictly degenerate) delay–scale direction; a formal Fisher/observability analysis is the planned next step. The decisive observation is that the metric-correct layout is the worse in-environment positioner *only* until the tag delay is corrected (after which the median ties at 72.9 mm), so the scale-biased layout's apparent advantage is conditional cancellation, not better physical calibration. A fair-comparison control with a single global tag-delay constant is still required.

The second contribution is methodological: a Vicon-grounded falsification protocol that separates metric correctness from same-environment positioning accuracy. It uses Sim(3) anchor-scale validation, a layout/delay transfer ablation, per-anchor range-bias decomposition with error-budget closure, phase-center sensitivity, and nested spatial CV, so that a single accuracy number cannot hide the mechanism.

As a discussion-level conjecture (not a contribution), a similar failure mode could occur wherever per-device bias and geometric scale share a weakly observable direction (GNSS clock–geometry, acoustic sound-speed–geometry); the analogy is imperfect because delay is additive while scale is multiplicative, so cancellation is only approximate over the working-distance range.

A raw-frame intervention that reduces the positive tag-anchor range tail reverses the ranking in favour of the metric-correct layout; this causal intervention is the decisive cancellation evidence and belongs in the manuscript main text.

4 Nearest Prior Art

Paper	What they did	How we differ
Hamer & D'Andrea (2018)	Built a self-calibrating one-way UWB network with clock synchronization and anchor computation for multi-robot localization.	We test whether the empirically best radio-only layout is physically correct, and show that a scale-biased layout can win in-environment (under an uncorrected tag delay) by cancelling structured range bias.
Lutz et al. (2019)	Used visual-inertial SLAM and fiducial markers to estimate UWB anchor poses and sensor error-model parameters.	Their calibration is aided by external visual structure; ours diagnoses a range-only self-calibration degeneracy using Vicon only as independent validation.
Ledergerber & D'Andrea (2018)	Estimated module-dependent range inaccuracies by maximum likelihood without external ground truth.	We show that fitting delay/range inaccuracies can reward a metric-incorrect (scale-biased) solution unless anchor metric scale is checked separately.
Luder et al. (2025) (AniTrack)	Reported that self-localized DWM3000 anchors gave better tag accuracy than laser-surveyed anchors (13.96 vs 16.57 cm average error).	They report an unexplained, small-sample paradox; we propose and Vicon-test a candidate mechanism via Sim(3) scale, a four-way layout/delay ablation, and common-mode scale recovery.
Pan et al. (2015) (GNSS)	Showed satellite-clock estimation can absorb orbit error and improve PPP positioning accuracy.	Their coupled-error absorption is a GNSS clock/orbit effect; our result is a UWB self-calibrated-layout scale bias that makes the metric-incorrect layout beat the correct one when the tag bias is uncorrected.

5 Risk and Status

Risk	Severity	One-line mitigation
One room and 24 static positions	High	State that transfer remains unproven and freeze the pipeline for second-room validation.
Cross-domain scale–delay ambiguity is prior art	Medium	Claim quantitative UWB characterization plus the in-environment cancellation result, not the bare ambiguity.
Common-mode scale recovery is a diagnostic	Medium	Present it as a mechanism re-parameterization, not as the deployed headline pipeline.
High-error tail remains	Medium	Separate median mechanism evidence from P95/dynamic performance claims.
Phase-center uncertainty	Low	Use sensitivity analysis and avoid treating the Vicon-oracle rank as the sole proof.

Status: 129 papers indexed, 37 cited; none we found explicitly characterizes the in-environment delay–layout cancellation, though De Preter (2019), Piavanini (2022), Batstone (2017), Hamer & D'Andrea (2018) and the acoustic/GNSS work are close and are engaged directly (a documented search protocol with query list and inclusion/exclusion criteria is included in the manuscript). Target: *Measurement* (IF $\sim$5.6). Working title: "Metric Correctness versus Positioning Accuracy in UWB Anchor Self-Calibration: A Vicon-Grounded Falsification Study."

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